

Luca Nigro

QUANTUM COMPUTING · QUANTUM INFORMATION · QUANTUM MACHINE LEARNING

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🇮🇹 Italian (Native) • 🇬🇧/🇺🇸 English (C2 - Proficient) • 🇪🇸 Spanish (B1 - Intermediate)



About Me

Quantum computing researcher (Ph.D. in Physics, University of Milan) developing quantum algorithms for optimization, simulation and quantum machine learning, and running them on real hardware: superconducting processors, neutral-atom arrays and quantum annealers. Author of peer-reviewed papers in *npj Quantum Information* and *Advanced Quantum Technologies*, with oral talks at APS Global Physics Summit, APS March Meeting and MDPI Entropy. Hands-on across the full stack, from theory and algorithm design to Qiskit, Cirq, PennyLane, Pulser, Bloqade and Ocean implementations and the Linux compute infrastructure behind them. Teach quantum computing to master's students and serve as peer reviewer and conference organizer.

Papers

- [1] Nigro, L., Monzani, F. & Prati, E. Non-markovian dissipation as a resource for quantum reservoir computing (2026). URL <https://arxiv.org/abs/2609.40172>. 2609.40172.
- [2] Nigro, L., Giorgi, G. L., Prati, E. & Zambrini, R. Analog neutral-atom for in-memory processing in quantum reservoir computing (2026). URL <https://arxiv.org/abs/2609.28441>. 2609.28441.
- [3] Ricci, E., Monzani, F., Nigro, L. & Prati, E. Quantum reservoir computing induced by controllable damping. *npj Quantum Information* (2026). URL <https://doi.org/10.1038/s41534-026-01229-8>.
- [4] Nigro, L., Sala, S., Amendola, A. & Prati, E. Leveraging quantum annealing for layout optimization. *Advanced Quantum Technologies* **8**, e00358 (2025). URL <https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/qute.202500358>.
- [5] Nigro, L., Barbieri, C. & Prati, E. Simulation of a three-nucleons system transition on quantum circuits. *Advanced Quantum Technologies* **8**, 2400371 (2025). URL <https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/qute.202400371>.
- [6] Monzani, F., Ricci, E., Nigro, L. & Prati, E. Non-unital noise in a superconducting quantum computer as a computational resource for reservoir computing (2025). URL <https://arxiv.org/abs/2409.07886>. 2409.07886.

Work Experience

PhD Researcher

UNIVERSITÀ DEGLI STUDI DI MILANO – QUANTUM INTELLIGENCE LAB

Milan, Italy

Oct 2023 - Oct 2026

- Quantum reservoir computing with engineered damping and non-Markovian dissipation.
- Wind farm layout optimization as a QUBO, benchmarked against classical solvers.
- Variational quantum simulation of nuclear transitions.
- Multi-platform implementation: Qiskit, Cirq, PennyLane, Ocean, Pulser, Bloqade.

Visiting Researcher

INSTITUTE FOR CROSS-DISCIPLINARY PHYSICS AND COMPLEX SYSTEMS (IFISC)

Palma de Mallorca, Spain

Sep 2025 - Feb 2026

- Analog neutral-atom quantum reservoir computing with Prof. Roberta Zambrini's group.
- Collaboration with international experts to strengthen technical expertise.

Teaching Assistant - Foundations of Quantum Computing

UNIVERSITÀ DEGLI STUDI DI MILANO

Milan, Italy

Feb 2024 - May 2026

- Built interactive Jupyter-based course material for master's students.
- Delivered hands-on lectures on multiple quantum frameworks and tutored master's theses.

Linux System Administrator & STEM Orientation Lecturer

UNIVERSITÀ DEGLI STUDI DI MILANO

Milan, Italy

Oct 2023 - Oct 2026

- Set up and maintained Linux workstations and shared computing resources for the research group.
- Delivered introductory quantum and AI lectures to high school students for STEM orientation.

Education

Università degli Studi di Milano

PHD IN PHYSICS

Topic: "Next generation of quantum algorithms for optimization and quantum chemistry"

Milan, Italy

Oct 2023 - Oct 2026

Università degli Studi di Milano

MSc IN PHYSICS

Thesis: "Simulation of Nuclear Reactions by a Gate Model Quantum Computer" (110/110 cum laude)

Milan, Italy

Oct 2020 - Jul 2023

University of Oslo

ERASMUS+

Studied Complex Media, Space Physics and Computational Physics

Oslo, Norway

Aug 2022 - Jan 2023

Università degli Studi di Milano-Bicocca

BSc IN PHYSICS

Thesis: "Study of HUBO Problems on Adiabatic Quantum Computers"

Milan, Italy

Oct 2016 - Sep 2020

Talks & Posters

Jul 2026	MDPI Entropy Conference (Oral) , "Quantum Reservoir Computing via Intrinsic and Engineered non-Unitary Dynamics"	Barcelona, ES
Mar 2025	APS Global Physics Summit (Oral) , "Benchmarking quantum computing for the wind farm layout optimization problem"	Anaheim, USA
Sep 2024	DICE Conference (Poster) , "Quantum digital simulation of the nuclear de-excitation of tritium"	Castiglioncello, Italy
Sep 2024	EQAI Summerschool (Poster) , "Enforcing fading memory of noisy quantum echo state networks"	Lignano Sab., Italy
Mar 2024	APS March Meeting (Oral) , "Quantum digital simulation of the nuclear de-excitation of tritium"	Minneapolis, USA
Dec 2023	CINECA HPC & QC Workshop (Oral) , "Hybrid quantum-classical algorithms for quantum simulation"	Bologna, Italy

Scientific Services

2026	Peer Reviewer , Quantum Machine Intelligence	Remote
2025	Master Thesis Tutor , MSc in Artificial Intelligence for Science and Technology	Milan, Italy
2025	Peer Reviewer , Quantum Information Processing	Remote
2025	Abstract Sorter , Joint APS March Meeting & April Meeting	Anaheim, CA, USA
2024	Peer Reviewer , Quantum Machine Intelligence	Remote
2024	Co-organizer & Chairperson , CINECA HPC&QC Workshop	Bologna, Italy
2024	Secretary , DICE24 - Quo Vadis, Fisica?	Castiglioncello, Italy

Skills

Programming	Python, Julia, C++, JavaScript, HTML, CSS
Quantum SDKs	Qiskit (IBM), Cirq (Google), PennyLane (Xanadu), Leap (D-Wave), Pulser (Pasqal), Bloqade (QuEra)
Tools & Tech	Git/GitHub, Linux (Bash), SSH, VS Code, Jupyter, $\LaTeX$, Markdown